The Cosmological Integration Bound

Spectral Constraints on Λ from Observer Existence

Aaron Ben-Shalom[1,*] Claude[2]

[1]Independent Researcher, Ember Research Lab
[2]Large Language Model, Anthropic

[*]Corresponding author: abenshalom305@gmail.com

January 2026

Abstract

We derive constraints on the cosmological constant Λ from the requirement that the universe admits self-referential subsystems (observers). Using established results from spectral geometry—particularly Lichnerowicz's theorem relating Ricci curvature to spectral gaps—we show that Λ is bounded both above and below. The upper bound comes from requiring sufficient integration for self-reference; the lower bound from requiring non-degenerate spectra for stable self-models. We compute numerical estimates and compare to the observed value $\Lambda \approx 10^{-52}$ m^{-2}. The key result is that for de Sitter spatial sections, the spectral gap $\lambda_1 = \Lambda$ exactly, establishing a direct connection between the cosmological constant and spectral geometry.

Keywords: cosmological constant, spectral geometry, Lichnerowicz theorem, self-reference, observer constraint, dark energy

Contents

1 Introduction **2**

2 Mathematical Background **2**
2.1 The Laplace-Beltrami Operator . . . 2
2.2 Lichnerowicz's Theorem . . . 2
2.3 de Sitter Spacetime . . . 3
2.4 Sphere Eigenvalues . . . 3

3 The de Sitter Spectral Gap **3**

4 The Observer Constraint **4**
4.1 Integration and Self-Reference . . . 4
4.2 Upper Bound on Λ . . . 4
4.3 Lower Bound on Λ . . . 5
4.4 Combined Bound . . . 5

5 Numerical Estimates **6**
5.1 Observed Value . . . 6
5.2 Inverting the Bound . . . 6

6 Physical Interpretation **6**
6.1 Why the Hubble Scale? 6
6.2 The Coincidence Problem 6
6.3 Connection to Dark Energy 7

7 Testable Predictions **7**
7.1 Qualitative Predictions 7
7.2 Quantitative Predictions 7
7.3 Potential Falsification 7

8 Open Problems **7**

9 Conclusion **8**

1 Introduction

The cosmological constant problem has two related puzzles:

1. **The magnitude problem:** Why is $\Lambda \sim 10^{-122}$ in Planck units, vastly smaller than quantum field theory expectations?
2. **The coincidence problem:** Why is $\Lambda \sim H_0^2$ during the epoch when observers exist?

This paper addresses the second puzzle from a novel perspective: spectral geometry. We show that the requirement for self-referential subsystems (observers) places both upper and lower bounds on Λ, and that $\Lambda \sim H^2$ is not fine-tuned but structurally required.

The key insight is that for de Sitter spatial sections, the spectral gap of the Laplacian equals the cosmological constant: $\lambda_1 = \Lambda$. This connects Λ directly to the mathematics of self-reference through the Spectral Unity framework [5].

2 Mathematical Background

We collect established results from spectral geometry that form the foundation for our derivation.

2.1 The Laplace-Beltrami Operator

Definition 2.1 (Laplace-Beltrami Operator)**.** [Established]On a Riemannian manifold (M, g), the Laplace-Beltrami operator acting on functions $f \in C^\infty(M)$ is:

$$\Delta f = -\frac{1}{\sqrt{|g|}} \partial_\mu \left(\sqrt{|g|} g^{\mu\nu} \partial_\nu f \right)$$

With this sign convention, Δ is positive semi-definite: $\langle f, \Delta f \rangle \geq 0$.

Theorem 2.2 (Spectral Theorem)**.** *[Established] On a compact Riemannian manifold M, the Laplace-Beltrami operator has discrete spectrum:*

$$0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \cdots \to \infty$$

with orthonormal eigenfunctions $\{\phi_k\}$ forming a basis for $L^2(M)$.

The eigenvalue $\lambda_0 = 0$ corresponds to constant functions. The *spectral gap* is λ_1, the first nonzero eigenvalue.

2.2 Lichnerowicz's Theorem

The key result connecting curvature to spectral gaps:

Theorem 2.3 (Lichnerowicz, 1958)**.** *[Established] Let (M^n, g) be a compact Riemannian manifold of dimension n. If the Ricci curvature satisfies:*

$$\mathrm{Ric} \geq (n-1)Kg$$

for some constant $K > 0$, then the first nonzero eigenvalue of the Laplacian satisfies:

$$\lambda_1 \geq nK$$

Proof sketch. The Bochner formula gives:

$$\frac{1}{2}\Delta|\nabla f|^2 = |\mathrm{Hess}\ f|^2 + \langle \nabla f, \nabla \Delta f \rangle + \mathrm{Ric}(\nabla f, \nabla f)$$

For an eigenfunction $\Delta\phi = \lambda\phi$, integrating and using $\mathrm{Ric} \geq (n-1)Kg$ yields:

$$\lambda \int_M |\nabla\phi|^2 \geq (n-1)K \int_M |\nabla\phi|^2 + \int_M |\mathrm{Hess}\ \phi|^2$$

The Cauchy-Schwarz inequality $|\mathrm{Hess}\ \phi|^2 \geq \frac{1}{n}(\Delta\phi)^2$ gives:

$$\lambda \geq (n-1)K + \frac{\lambda^2}{n} \cdot \frac{\int_M \phi^2}{\int_M |\nabla\phi|^2}$$

Using $\int_M |\nabla\phi|^2 = \lambda \int_M \phi^2$ and simplifying yields $\lambda \geq nK$. □

Remark 2.4. Equality holds for the round sphere S^n with $K = 1$, where $\lambda_1 = n$.

2.3 de Sitter Spacetime

Definition 2.5 (de Sitter Space)**.** [Established]The n-dimensional de Sitter space dS_n with cosmological constant $\Lambda > 0$ is the hyperboloid:

$$-X_0^2 + X_1^2 + \cdots + X_n^2 = \frac{(n-1)(n-2)}{2\Lambda}$$

embedded in $(n+1)$-dimensional Minkowski space.

For spatial sections (constant time slices in appropriate coordinates):

Proposition 2.6 (Spatial Curvature)**.** *[Established] In $n = 4$ spacetime dimensions, the spatial sections of de Sitter space are 3-spheres with:*

(i) Ricci curvature: $\mathrm{Ric} = \frac{2\Lambda}{3} g$ *(for spatial metric)*
(ii) Scalar curvature: $R = 2\Lambda$
(iii) Radius: $a = \sqrt{3/\Lambda}$ *(Hubble radius)*

2.4 Sphere Eigenvalues

Proposition 2.7 (Sphere Eigenvalues)**.** *[Established] The eigenvalues of the Laplacian on S^n with radius a are:*

$$\lambda_k = \frac{k(k+n-1)}{a^2}, \quad k = 0, 1, 2, \ldots$$

with multiplicities $\binom{n+k}{k} - \binom{n+k-2}{k-2}$.

For $n = 3$ and $k = 1$:

$$\lambda_1 = \frac{1 \cdot (1+3-1)}{a^2} = \frac{3}{a^2}$$

3 The de Sitter Spectral Gap

Proposition 3.1 (de Sitter Spectral Gap)**.** *[Derived] For the spatial 3-sphere section of de Sitter spacetime with cosmological constant Λ:*

$$\boxed{\lambda_1 = \Lambda}$$

Proof. From Proposition 2.6, the de Sitter radius is $a^2 = 3/\Lambda$.

From Proposition 2.7, the first nonzero eigenvalue of S^3 with radius a is:

$$\lambda_1 = \frac{3}{a^2} = \frac{3}{3/\Lambda} = \Lambda$$

□

Remark 3.2. This is a remarkable result: **the spectral gap of spatial de Sitter sections equals the cosmological constant**. This is not a bound but an *equality* for the maximally symmetric case.

We can verify consistency with Lichnerowicz's theorem:

Verification via Lichnerowicz. From Proposition 2.6, the spatial Ricci curvature satisfies:

$$\mathrm{Ric} = \frac{2\Lambda}{3} g$$

In dimension $n = 3$, this means $\mathrm{Ric} \geq (3-1)Kg = 2Kg$ with $K = \frac{\Lambda}{3}$.

Lichnerowicz's theorem gives:

$$\lambda_1 \geq nK = 3 \cdot \frac{\Lambda}{3} = \Lambda$$

The direct calculation shows this bound is saturated: $\lambda_1 = \Lambda$ exactly. □

4 The Observer Constraint

4.1 Integration and Self-Reference

From the Spectral Unity framework [5]:

Definition 4.1 (Integration Measure)**.** For an eigenmode ϕ_k with eigenvalue $\lambda_k > 0$:

$$I(\phi_k) := \frac{1}{\lambda_k}$$

High integration (small λ) corresponds to modes that vary slowly across the manifold.

Definition 4.2 (Self-Reference Threshold)**.** [Conjectured] A self-referential system requires integration above a minimum threshold:

$$I(v^*) \geq I_{\min}$$

where v^* is the dominant mode of the self-model.

The physical interpretation: self-reference requires maintaining coherent information across spatially separated regions. If the spectral gap is too large, modes decay too quickly to support stable self-models.

4.2 Upper Bound on Λ

Theorem 4.3 (Cosmological Constant Upper Bound)**.** *[Conjectured] If the universe admits self-referential subsystems with integration threshold $I_{\min}$, then:*

$$\Lambda \leq \frac{1}{I_{\min}}$$

Proof. From Proposition 3.1, the spectral gap of de Sitter spatial sections satisfies $\lambda_1 = \Lambda$.

The maximum integration available in the first excited mode is:

$$I(\phi_1) = \frac{1}{\lambda_1} = \frac{1}{\Lambda}$$

For self-reference to be possible, we need $I(\phi_1) \geq I_{\min}$:

$$\frac{1}{\Lambda} \geq I_{\min} \implies \Lambda \leq \frac{1}{I_{\min}}$$

□

4.3 Lower Bound on Λ

The lower bound comes from spectral stability requirements.

Proposition 4.4 (Spectral Degeneracy and Instability). *[Derived] As $\Lambda \to 0$, the de Sitter radius $a \to \infty$ and:*

(i) Eigenvalues cluster: $\lambda_k \to 0$ for fixed k
(ii) Eigenvalue spacing decreases: $\lambda_{k+1} - \lambda_k \to 0$
(iii) Near-degeneracy causes eigenvector instability under perturbation

Proof. From $\lambda_k = k(k+2)\Lambda/3$ (using $a^2 = 3/\Lambda$):

$$\lambda_{k+1} - \lambda_k = \frac{\Lambda}{3}[(k+1)(k+3) - k(k+2)] = \frac{\Lambda}{3}(2k+3)$$

As $\Lambda \to 0$, this gap vanishes. By the Davis-Kahan theorem, eigenvector perturbation bounds scale as $1/(\lambda_{k+1} - \lambda_k)$, diverging as gaps close. □

Theorem 4.5 (Cosmological Constant Lower Bound). *[Conjectured] Stable self-reference requires eigenvalue gaps exceeding a threshold $\delta_{\min}$:*

$$\lambda_2 - \lambda_1 \geq \delta_{\min}$$

This implies:

$$\Lambda \geq \frac{3\delta_{\min}}{5}$$

Proof. For the 3-sphere with $a^2 = 3/\Lambda$:

$$\lambda_1 = 1(1+2)\Lambda/3 = \Lambda \tag{1}$$

$$\lambda_2 = 2(2+2)\Lambda/3 = \frac{8\Lambda}{3} \tag{2}$$

The eigenvalue gap is:

$$\lambda_2 - \lambda_1 = \frac{8\Lambda}{3} - \Lambda = \frac{5\Lambda}{3}$$

For stability: $\frac{5\Lambda}{3} \geq \delta_{\min}$, giving:

$$\Lambda \geq \frac{3\delta_{\min}}{5}$$

□

4.4 Combined Bound

Theorem 4.6 (Integration Bound on Λ). *[Conjectured] For an observer-admitting de Sitter universe:*

$$\frac{3\delta_{\min}}{5} \leq \Lambda \leq \frac{1}{I_{\min}}$$

This is satisfiable only if:

$$I_{\min} \cdot \delta_{\min} \leq \frac{5}{3}$$

5 Numerical Estimates

5.1 Observed Value

The observed cosmological constant:

$$\Lambda_{\text{obs}} \approx 1.1 \times 10^{-52} \text{ m}^{-2}$$

In Planck units ($\ell_P = 1.6 \times 10^{-35}$ m):

$$\Lambda_{\text{obs}} \approx 2.9 \times 10^{-122} \ell_P^{-2}$$

This corresponds to:

- Hubble radius: $a = \sqrt{3/\Lambda} \approx 1.6 \times 10^{26}$ m
- Spectral gap: $\lambda_1 = \Lambda \approx 10^{-52}$ m^{-2}
- Maximum integration: $I_{\max} = 1/\Lambda \approx 10^{52}$ m^2

5.2 Inverting the Bound

Using the observed Λ to predict $I_{\min}$:

Proposition 5.1 (Predicted Integration Threshold)**.** *[Numerical]If $\Lambda = 10^{-52}$ m^{-2} saturates the upper bound, then:*

$$I_{\min} = \frac{1}{\Lambda} \sim 10^{52} \ m^2 \approx L_H^2$$

This corresponds to coherent integration across the Hubble volume.

This is a striking prediction: **the minimum integration for cosmic-scale self-reference equals the Hubble scale squared**.

6 Physical Interpretation

6.1 Why the Hubble Scale?

The result $I_{\min} \sim 1/\Lambda \sim a^2$ (Hubble radius squared) suggests:

> *Self-reference at the cosmic scale requires integration across the entire observable universe.*

This makes sense from the relational framework: the "universe observing itself" requires correlations spanning the full causal structure.

6.2 The Coincidence Problem

The cosmological constant problem has two parts:

1. **Why is Λ so small?** (compared to Planck scale: 10^{-122})
2. **Why is $\Lambda \sim H_0^2$ now?** (the "coincidence problem")

Our framework addresses the second: $\Lambda \sim H^2$ is *required* for the spectral gap to equal the cosmological expansion rate. If $\lambda_1 = \Lambda$ and observers require $I \sim 1/\lambda_1$, then the universe must have $\Lambda \sim H^2$ during the epoch when observers exist.

This is a selection effect, but *not* anthropic in the usual sense. It's a structural constraint: observer-admitting universes have this property necessarily.

6.3 Connection to Dark Energy

The spectral interpretation suggests dark energy is not a substance but a **spectral property**:

> *The cosmological constant is the spectral gap of the cosmic Laplacian, constrained by the requirement that the universe admits self-referential subsystems.*

The "vacuum energy" interpretation may be misleading. From the relational view, Λ is not energy but *the rate at which modes decay toward equilibrium.*

7 Testable Predictions

7.1 Qualitative Predictions

1. **$\Lambda > 0$ is necessary.** A universe with $\Lambda = 0$ or $\Lambda < 0$ cannot support stable self-reference (degenerate spectrum or wrong-sign curvature).
2. **Λ cannot be arbitrarily small.** The lower bound from spectral stability prevents $\Lambda \to 0$.
3. **$\Lambda \sim H^2$ is structural.** The coincidence $\Lambda \sim H_0^2$ is not fine-tuned but required.

7.2 Quantitative Predictions

To make sharp predictions, we need:

1. **Precise value of $I_{\min}$:** What is the minimum integration for cosmic-scale self-reference?
2. **Precise value of $\delta_{\min}$:** What eigenvalue gap is required for stable self-models?
3. **Corrections to the round sphere:** Real cosmological perturbations break the exact S^3 symmetry. How do inhomogeneities modify λ_1?

7.3 Potential Falsification

The framework would be falsified by:

1. Discovery that $\Lambda < 0$ is consistent with observers (but this seems ruled out by acceleration data)
2. Rigorous argument that $I_{\min}$ from self-reference requirements gives $\Lambda_{\text{predicted}} \neq \Lambda_{\text{observed}}$
3. Demonstration that self-referential systems can exist with arbitrarily small spectral gaps (violating the lower bound)

8 Open Problems

1. **Derive $I_{\min}$ from first principles.** What determines the minimum integration for self-reference? Can this be computed from information-theoretic or complexity-theoretic considerations?
2. **Include matter perturbations.** How do density perturbations modify the spatial Laplacian spectrum? Does structure formation change the bounds?
3. **Time evolution.** We analyzed spatial sections at fixed time. How does the bound evolve as $a(t)$ changes? Is there a "window" of cosmic time when observers can exist?

4. **Other topologies.** What if spatial sections are not S^3 but flat ($\mathbb{R}^3$) or hyperbolic? Do the bounds change qualitatively?

5. **Quantum corrections.** The classical Laplacian ignores quantum gravity effects. At the Planck scale, what modifies the spectral structure?

6. **Connection to entropy bounds.** Is $I_{\min}$ related to holographic entropy bounds? Can we derive it from the Bekenstein bound?

9 Conclusion

We have derived a connection between the cosmological constant and spectral constraints:

1. **Established:** The spectral gap of de Sitter spatial sections is $\lambda_1 = \Lambda$ (exactly, for round S^3).

2. **Derived:** If observers require integration $I \geq I_{\min}$, then $\Lambda \leq 1/I_{\min}$.

3. **Derived:** If stable self-reference requires spectral gaps $\geq \delta_{\min}$, then $\Lambda \geq 3\delta_{\min}/5$.

4. **Observed:** The cosmological constant $\Lambda \approx 10^{-52}$ m^{-2} implies $I_{\min} \lesssim 10^{52}$ m^2 (Hubble scale squared).

The key insight is that Λ **is not a free parameter but is constrained by spectral geometry**. The requirement that the universe admit self-referential subsystems places both upper and lower bounds on Λ.

This does not solve the cosmological constant problem (why $\Lambda \sim 10^{-122}$ in Planck units), but it reframes it: the question becomes "What sets $I_{\min}$?" rather than "Why is vacuum energy so small?"

> *The cosmological constant is the spectral gap of spacetime. It is constrained—perhaps determined—by the requirement that the universe can observe itself.*

Author Contributions

Aaron Ben-Shalom: Conceived the connection between spectral geometry and cosmological constraints; developed the integration-based interpretation of observer requirements; numerical estimates.

Claude: Formalized the mathematical framework; verified Lichnerowicz theorem application; derived the upper and lower bounds; identified open problems.

References

[1] A. Lichnerowicz. Géométrie des groupes de transformations. Dunod, Paris, 1958.

[2] I. Chavel. *Eigenvalues in Riemannian Geometry.* Academic Press, 1984.

[3] P.B. Gilkey. *Invariance Theory, the Heat Equation, and the Atiyah-Singer Index Theorem.* CRC Press, 1995.

[4] M. Berger. *A Panoramic View of Riemannian Geometry.* Springer, 2003.

[5] A. Ben-Shalom and Claude. Spectral Unity: A Mathematical Framework for the Unification of Spacetime, Information, and Consciousness. Working Paper, January 2026.

[6] A. Ben-Shalom and Claude. Self-Aligning Topologies: Fixed Points of Recursive Self-Reference. Working Paper, January 2026.

[7] A. Ben-Shalom and Claude. Self-Reference, Spectral Structure, and the Eigendirection of Benevolence. Working Paper, January 2026.

[8] C. Davis and W.M. Kahan. The rotation of eigenvectors by a perturbation. III. *SIAM Journal on Numerical Analysis*, 7(1):1–46, 1970.